Partially Fault-Tolerant Quantum Computation for Megaquop Applications

Where partial fault-tolerance pays off, and where that stops



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Megaquop Machines and PFTQC

- NISQ era
- ~100 qubits
- Error mitigation required

- Megaquop machines (Preskill, 2025)
- 10^4 to 10^5 qubits
- Run 10^6 logical operations with an error of 10^{-6} per gate

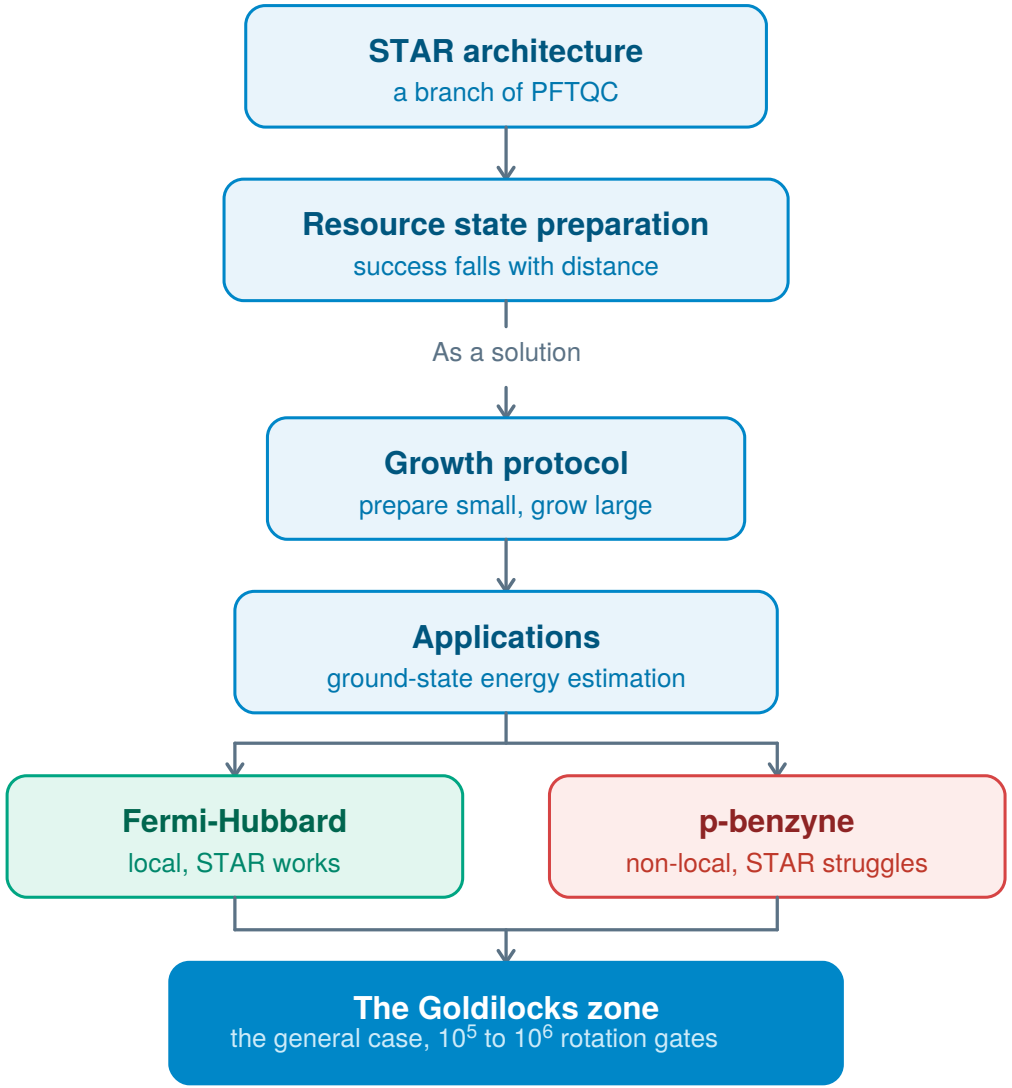
- Full FTQC era
- More than a million qubits
- Fault-tolerant gate operations

Partial fault-tolerance (Akahoshi et al., 2024; Toshio et al., 2025):

- Clifford gates applied fault-tolerantly, by lattice surgery
- Resource states (small angle Z rotations) are prepared non-fault-tolerantly
- Error mitigation is required to absorb residual errors

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From Architecture to Applications



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STAR Architecture

Full FTQC runs on Clifford + T gates.

For a small angle:

$$R_Z(\theta) \;\rightarrow\; \sim 100\ T \text{ gates} \;\rightarrow\; \text{large qubit overhead}$$

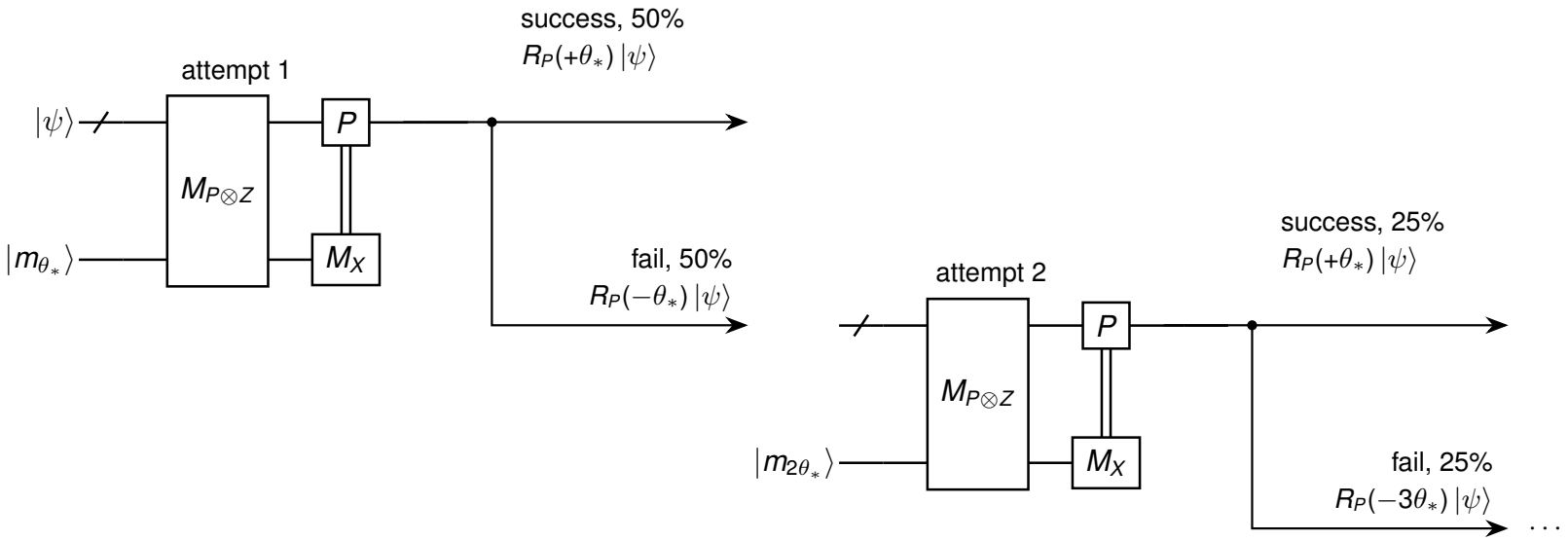
The STAR architecture

Prepare a resource state in an analog way! So no synthesis is needed.

$$|m_{\theta_*}\rangle = \cos\theta_* |+\rangle_L + i \sin\theta_* |-\rangle_L$$

Then, teleport the resource state into the target qubit.

Repeat until success (RUS): on average **two attempts**, so about two resource states per rotation instead of **100 *T* gates**



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Resource State Preparation

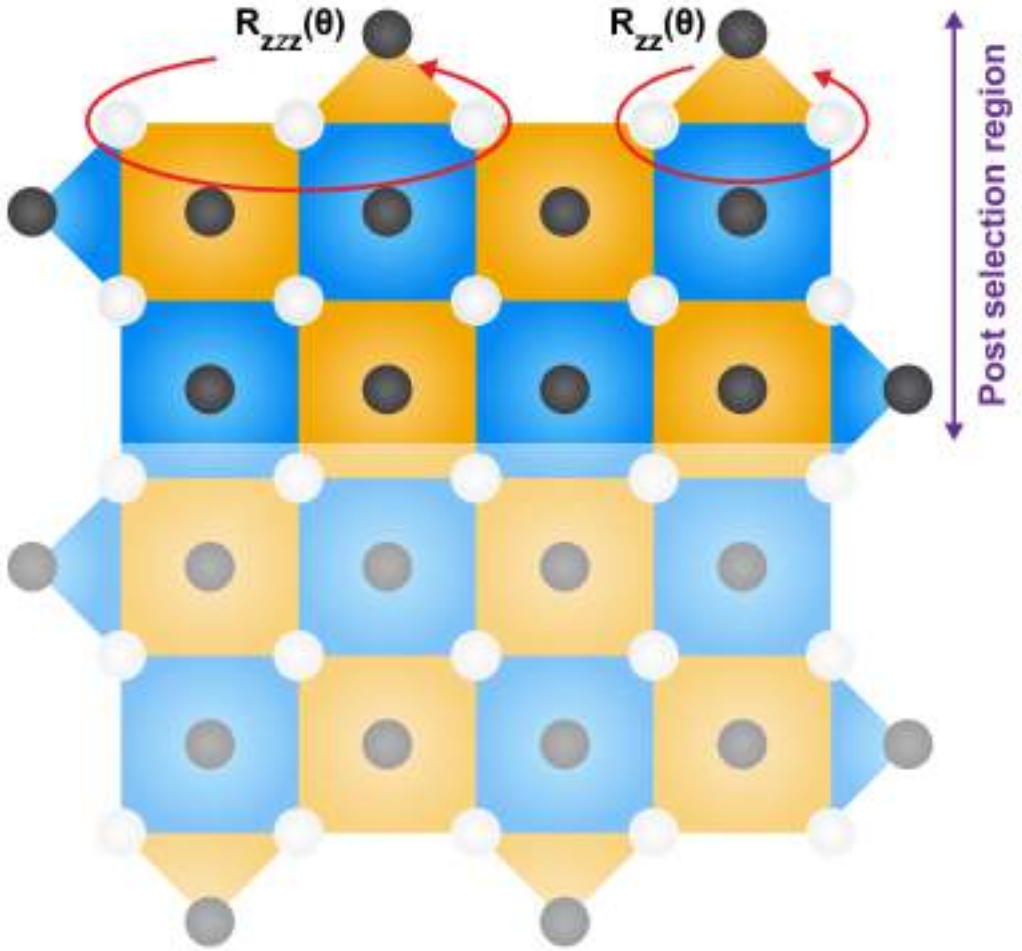
Preparing a resource state $|m_{\theta_*}\rangle$

1. **Prepare a** $|+\rangle_L$ in the codespace, where the stabilizers are $+1$.

2. **Apply the physical rotation gates** along the top row, which takes the state to a superposition of stabilizer states.

3. **Perform two additional rounds** of syndrome extraction, post-selecting on all three rounds, which brings the patch back to the codespace!

Protocol and simulation methods follow **Toshio et al., 2025**.



A distance 5 rotated surface code. Orange faces represent X stabilizers, blue faces represent Z stabilizers, and the top row of physical qubits supports the Z_L operator.

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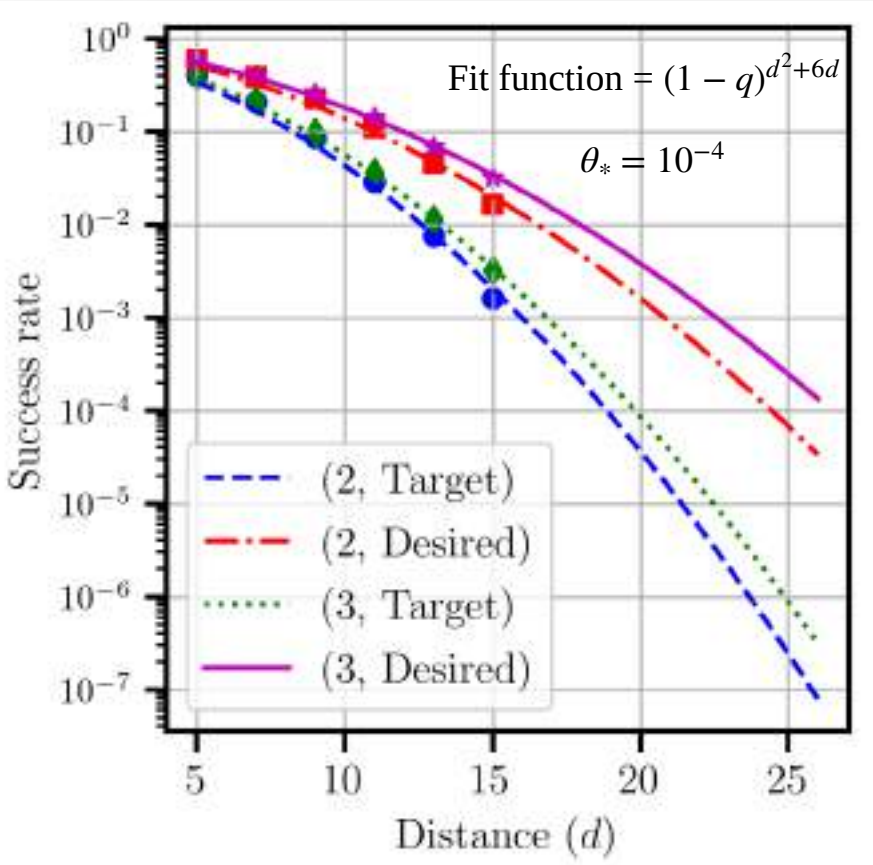
Realistic Hardware Parameters

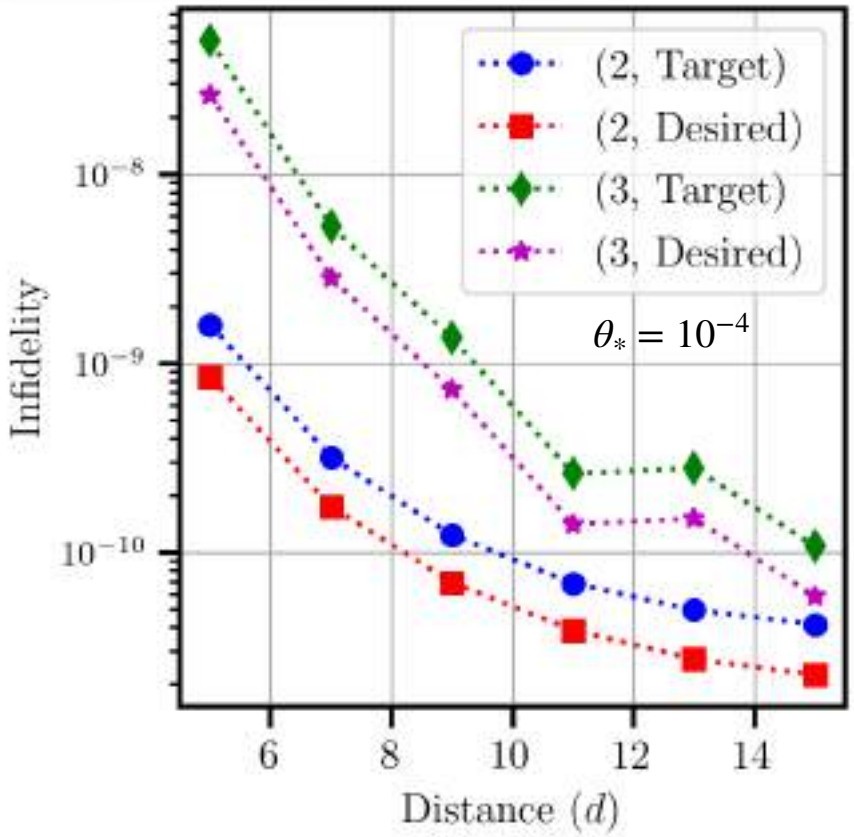
- We follow the same protocol and simulation method, but with a realistic noise model.
- It is a full circuit-level noise model with two superconducting hardware specifications (Mohseni et al., 2024), rather than a uniform depolarizing channel.

Hardware Parameter	Target	Desired
T1, T2 times	**200 µs**	**340 µs**
Single-qubit gate error	**0.0002**	**0.00012**
Two-qubit gate error	**0.0005**	**0.00029**
State preparation error	**0.01**	**0.00588**
Measurement error	**0.005**	**0.00294**
Reset error	**0.005**	**0.00294**
Error suppression rate Λ	**9.3**	**18**

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Success Rate and Infidelity of Resource State Preparation





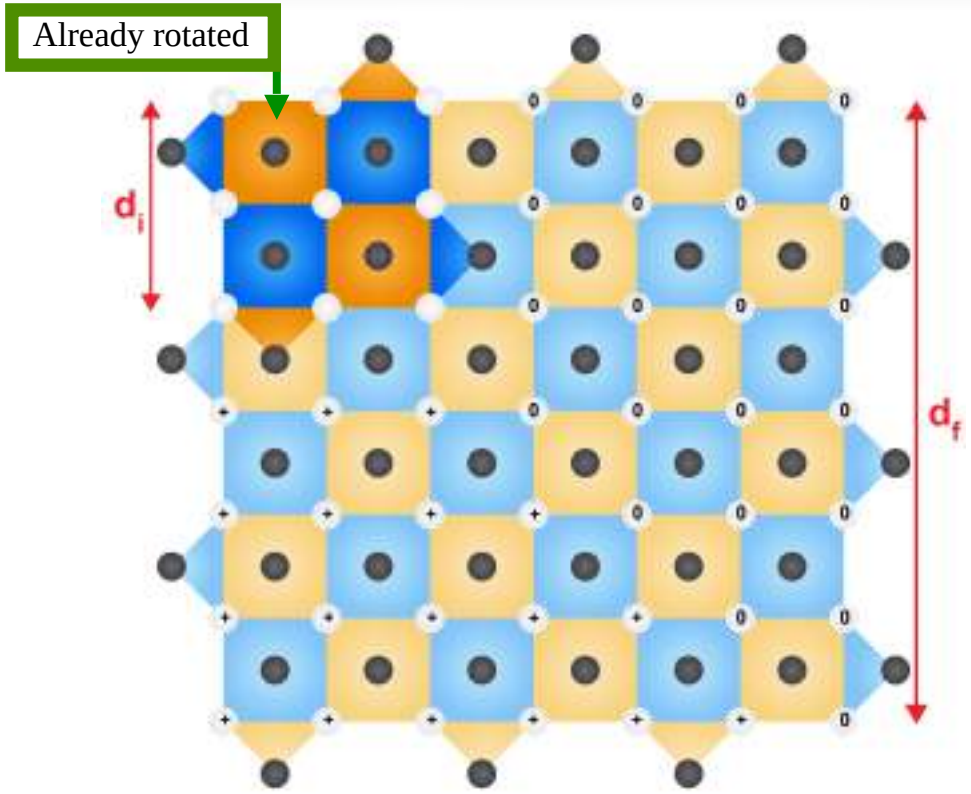
- At $d = 13$, the success is below 1% on target hardware
- Below 1%, a resource state costs more than the 100 T gates full FTQC would require

- Physical Z-errors cannot be detected if coincident with physical rotation gates
- Logical infidelity is given by $P_L = P_{ud}\theta_*^{2-\frac{2}{k}}$

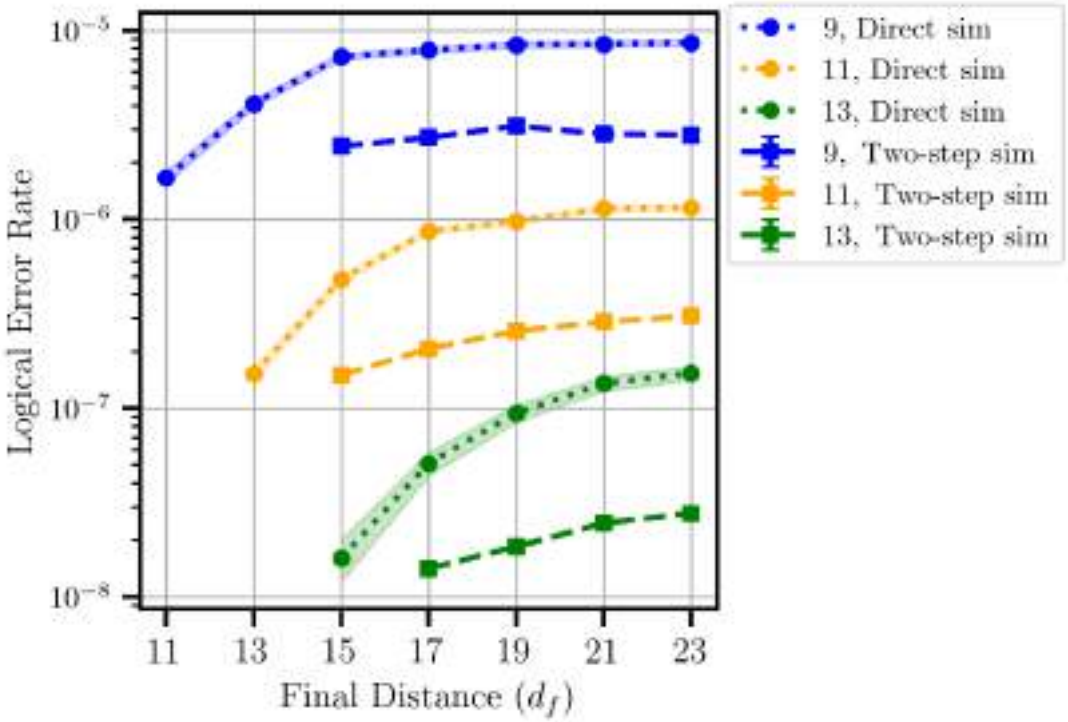
The infidelity stays in the megaquop regime, but the success rate drops sharply with distance.

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Growth Protocol



- The patch at d_i is expanded, with new data qubits prepared in the zero and plus states (Fowler and Gidney, 2018).
- Then, d_f rounds of error correction are performed.



- Larger d_i result in lower infidelity
- Direct growth: $d_i \rightarrow d_f$
- Two-step growth: $d_i \rightarrow d_i + 2 \rightarrow d_f$
- Two-step growth is slower ($d_i + 2$ more rounds)

Growth gives us the larger patch we need, but could result in greater infidelity.

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Optimal Initial Distance

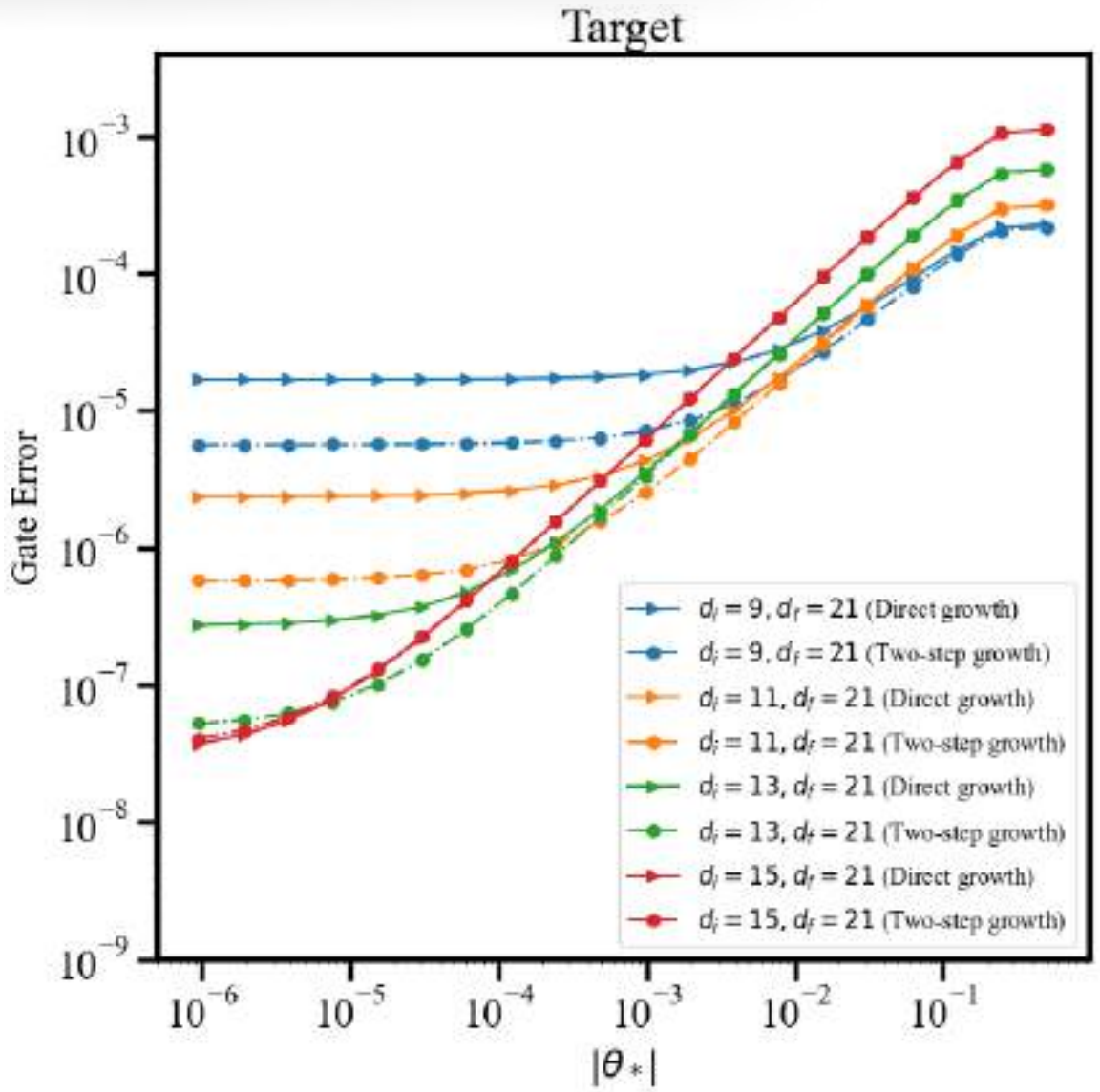
Two sources of error

1. Analog rotation error grows with the angle.
2. Growth error is independent of the angle, and falls with d_i:

$$P_L(\theta_*) \sim O(\theta_*) + 2I_{\text{growth}}(d_i)$$

They compete, so each angle has its own best d_i.

- The residual error **is not corrected; it is mitigated**.
- Probabilistic error cancellation (PEC) cancels out what is left.
- The cost is sampling overhead: $\sim \exp\left(\sum P_L\right)$
- The sum runs over every rotation in the circuit.



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Applications

The problem we aim to solve

- Ground-state energy estimation:
  - The *2D Fermi–Hubbard* model, an $L \times L$ lattice of interacting electrons
  - *p-benzyne*, with active spaces from 6 to 26 orbitals, or 12 to 52 logical qubits. The spaces are chosen to test the estimation pipeline, not to claim quantum advantage.

With which algorithm

QCELS (Ding and Lin, 2023):

- Shallower circuits than standard phase estimation
- Trotterized time evolution, which produces the small angles where the STAR architecture is useful

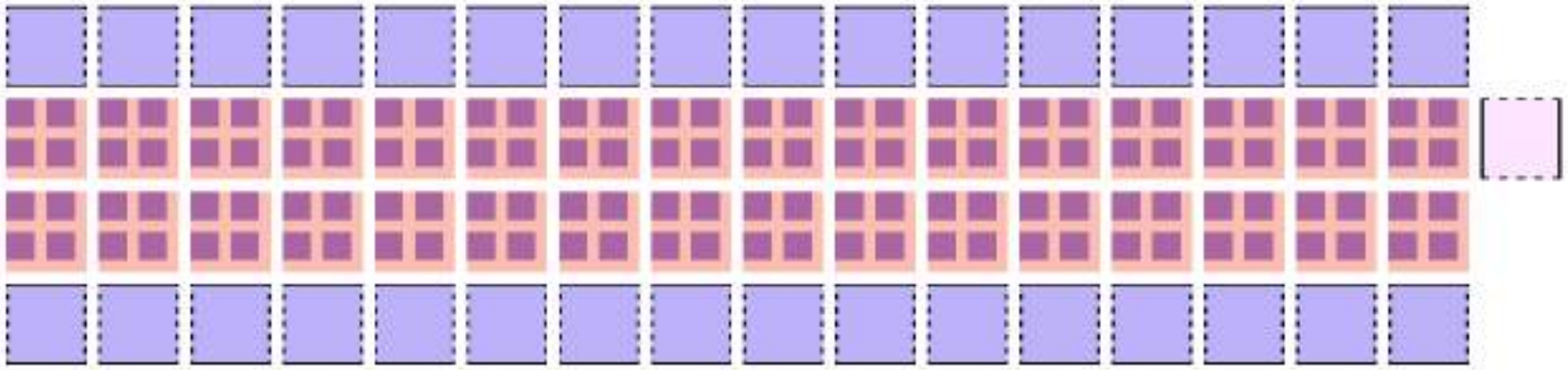
To what level of accuracy

- p-benzyne: 1.6 mHa, chemical accuracy
- Fermi–Hubbard model: $0.005L^2$, which loosens with L; and 0.01 fixed

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2D Fermi–Hubbard Model

- An example of an FH model where $L = 4$:
  - Each column represents one site
  - Particles with spin up are encoded in the top row and ones with spin down are encoded in the bottom row



$\varepsilon = 0.005L^2$

L	STAR qubits	FTQC qubits	STAR time	Slowdown	FTQC total
8	2.3E+05	8.9E+05	5.0 min	1.7	21 s
10	3.5E+05	1.1E+06	3.8 min	1.8	18 s

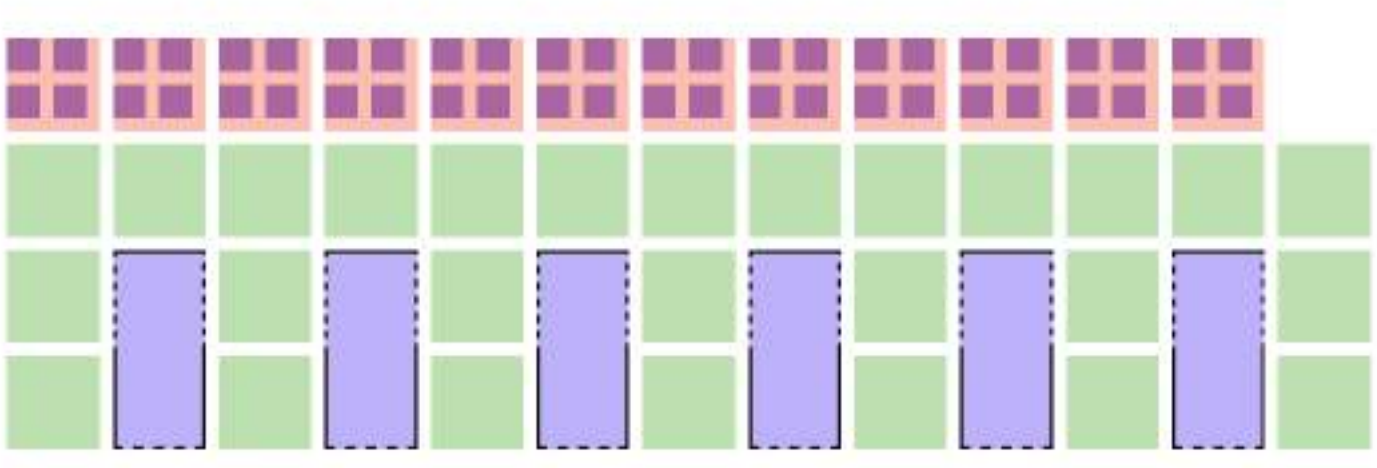
- FTQC estimates generated using TopQAD* with plaquette Trotterization
- “STAR time” does not include the PEC sampling overhead
- “Slowdown” is the extra factor needed to include PEC sampling overhead

* *TopQAD* is 1QBit’s Topological Quantum Architecture Design software suite (topqad.1qbit.com)

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p-benzyne

- p-benzyne, at active spaces from 6 to 26 orbitals
- Hamiltonian is non-local → rotations run one at a time → deeper circuit



Violet are data, green are bus, pink is the factory

Orbitals	STAR qubits	FTQC qubits	STAR time	FTQC time	Slowdown
6	2.9E+04	1.4E+06	2.6 min	43 s	2.5
18	1.5E+05	3.6E+06	14.4 h	6.9 h	10^{10}
26	2.5E+05	4.7E+06	4.4 days	1.2 days	10^{51}

FTQC estimates generated using TopQAD with double-factorized qubitization

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The Goldilocks Zone

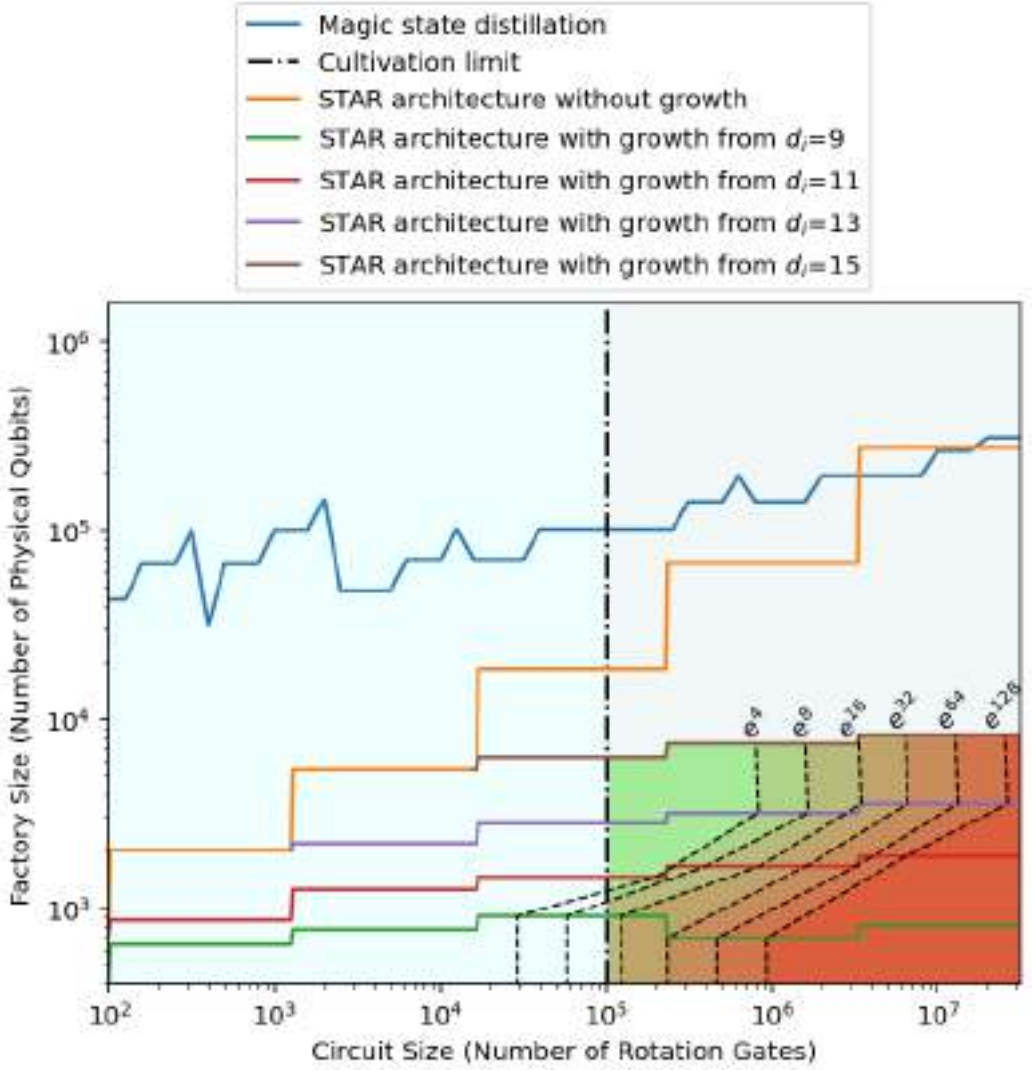
No Hamiltonian is involved

- Fix the circuit at 100 logical qubits, angles at 10^{-4}, error budget at 1%
- As the rotation count grows, how much does the factory increase?
- Which of distillation, cultivation and STAR performs best?

Results can be grouped into three regimes

- **Below 10^5, cultivation** is least expensive
- **Between 10^5 and 10^6, the STAR architecture** has the smallest factory
- **Above 10^6**, the slowdown makes the STAR architecture impractical

Generated using TopQAD, with $\theta_* = 10^{-4}$



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Conclusions and Future Research

- **With realistic superconducting hardware, the success rate decreases rapidly as the code distance of the target patch increases.**

- **The growth protocol brings the patch to the distance the core processor needs, but injects additional error.**
  - **The two-step growth protocol has a lower infidelity, but takes longer.**
  - **Techniques borrowed from magic state cultivation could help.**

- **Significant PEC sampling overheads can make the STAR architecture impractical.**
  - **Better error mitigation strategies are needed.**

- **Goldilocks scale: the STAR architecture is most effective on the scale of 10^5 to 10^6 small-angle rotations.**

Thank You!

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